

Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_best_single_fixed_heuristic, phase8_full_solver_evolution, phase8_llm_evolved_adaptive_controller, phase8_llm_static_selector.

Run Metadata

- run_name: run_20260520_195202_h
- started_at_local: 2026-05-20 19:52:02
- finished_at_local: 2026-05-20 20:27:58
- duration_hhmm: 00:36
- duration_seconds: 2156.549
- seed_offset: 7000
- replicate_label: h
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1463.164843	0.07226	0.007284	0.03122	0.0	0.0	True
phase8_random_portfolio	random_portfolio	none	0.189641	0.117381	1493.831657	0.193616	0.038767	0.094352	0.0	0.0	False
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1490.476029	0.073609	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLI B Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1521.140557	0.075123	0.007284	0.031222	0.0	0.0	False
phase8_llm_static_selector	static_selector	none	0.229271	0.157011	273.494286	0.229271	0.068101	0.127479	0.0	0.62	True
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.214311	0.142051	391.322871	0.214311	0.045168	0.107484	0.772579	2.1175	True
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.120261	0.048001	7723.547714	0.634817	0.001857	0.045479	0.722823	0.62	False
phase8_full_solver_evolution	full_solver	none	0.07918	0.00692	509.750157	0.07918	0.010298	0.035675	0.548573	0.76	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1463.164843 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.189641, selector regret 0.117381, runtime-adjusted gap 0.193616.
- Family transfer 0.038767 and combined transfer 0.094352.
- Runtime 1493.831657 ms, runtime inflation 0.020959, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.073609.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 1490.476029 ms, runtime inflation 0.018666, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.075123.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1521.140557 ms, runtime inflation 0.039624, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.229271, selector regret 0.157011, runtime-adjusted gap 0.229271.
- Family transfer 0.068101 and combined transfer 0.127479.
- Runtime 273.494286 ms, runtime inflation -0.81308, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: annealed_multi_start:cluster_first_local_search:edge_preserving_restart:annealed_multi_start:stag3:cand2:restart1.
- Controller rule summary: {"acceptance_bias": 0.005, "candidate_limit_offset": 2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic, instance-conditioned frozen-portfolio selector that favors farthest-insertion+2opt generally, then switches to cluster/bottleneck/grid/corridor aware constructions. Includes bounded online-style tuning parameters for the selected schedule but no", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.08, "name": "det_tsp_portfolio_oracle_approach_static_v1", "nearest_neighbor_trap_heuristic": "candidate_pruned_two_opt", "random_like_heuristic": "annealed_multi_start", "restart_offset": 1, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.1, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.045168 and combined transfer 0.107484.
- Runtime 391.322871 ms, runtime inflation -0.73255, mean novelty 0.772579, and complexity 2.1175.
- Pareto efficient: True.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:edge_preserving_restart:nearest_neighbor_multistart:stag3:cand1:restart0.
- Controller rule summary: {"acceptance_bias": 0.007, "candidate_limit_offset": 1, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive scheduler over a frozen heuristic portfolio with bounded online switching and interpretable tuning offsets.", "failed_perturbation_threshold": 3, "finalization_policy": {"determinism": "All choices are deterministic functions of instance descriptors and current online

state; no random sampling is performed by this controller.", "rule": "At the end of budget, or after any controller switch, do only one final compact refinement consistent with the currently selected heuristic; do not allow additional controller-level switches in the same decision step.", "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.09, "name": "tsp_hh_oracle_portfolio_transfer_v3", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 0, "selection_rules": [{"initial_construction_choice": [{"choose": "edge_preserving_restart", "when": "bottleneck_score >= 0.58 or two_cluster_bottleneck_score >= 0.48"}, {"choose": "cluster_first_local_search", "when": "clustering_score >= 0.56 and cluster_separation >= 0.30 and convex_hull_ratio >= 0.88"}, {"choose": "candidate_pruned_two_opt", "when": "grid_likeness >= 0.30"}, {"choose": "limited_three_opt", "when": "corridor_score >= 0.45 or elongated_structure >= 0.44"}, {"choose": "annealed_multi_start", "when": "nearest_neighbor_trap_score >= 0.34 or nn_distance_cv >= 0.68"}, {"choose": "farthest_insertion_two_opt", "when": "True"}], "instance_condition_tuning": [{"action": "candidate_limit_offset = clamp(candidate_limit_offset + 2, -6, 6) to tighten pruning for grid-like structure.", "when": "grid_likeness >= 0.30"}, {"action": "candidate_limit_offset = clamp(candidate_limit_offset + 1, -6, 6) to avoid over-pruning on non-grid instances.", "when": "grid_likeness < 0.14 and candidate_limit_offset < 1"}, {"action": "restart_offset = clamp(restart_offset + 1, -3, 4) to increase restart pressure on bottleneck/two-cluster layouts.", "when": "bottleneck_score >= 0.60"}, {"action": "restart_offset = clamp(restart_offset - 1, -3, 4) to reduce excessive restart churn on well-mixed instances.", "when": "bottleneck_score < 0.35"}, {"action": "temperature_scale = clamp(temperature_scale * 0.98, 0.5, 1.8); acceptance_bias = clamp(acceptance_bias + 0.001, -0.01, 0.02) to stabilize on well-shaped clustered instances.", "when": "clustering_score >= 0.56 and convex_hull_ratio >= 0.88"}, {"action": "temperature_scale = clamp(temperature_scale * 1.10, 0.5, 1.8); acceptance_bias = clamp(acceptance_bias + 0.003, -0.01, 0.02) to broaden acceptance under higher distance variance.", "when": "distance_cv >= 0.52 and distance_mean_norm >= 0.25"}, {"action": "candidate_limit_offset = clamp(candidate_limit_offset + 1, -6, 6) to exploit stable nearest-neighbor structure.", "when": "nn_distance_variance_norm <= 0.0012"}, {"action": "temperature_scale = clamp(temperature_scale * 1.06, 0.5, 1.8); acceptance_bias = clamp(acceptance_bias + 0.001, -0.01, 0.02) to escape trap-like layouts.", "when": "nearest_neighbor_trap_score >= 0.34"}, {"action": "acceptance_bias = clamp(acceptance_bias + 0.001, -0.01, 0.02) to nudge exploration during mild stagnation.", "when": "stagnation_length >= 2 and edge_length_skew <= 0.55"}], "online_scheduling_choice": [{"phase": "early", "rule": "if time_budget_used < time_budget_trigger: keep current heuristic; only allow its built-in local improvement (no controller switch unless failed perturbation triggers)."}, {"phase": "failed_perturbation_escape", "rule": "if failed_perturbation_count >= failed_perturbation_threshold: switch to random_like_heuristic (one switch max per decision step)."}, {"phase": "stagnation_recovery", "rule": "if stagnation_length >= stagnation_threshold and recent_improvement_rate <= low_improvement_threshold and edge_length_skew <= 0.66: switch to stagnation_switch_heuristic."}, {"phase": "time_pressure", "rule": "if time_budget_used >= time_budget_trigger and (recent_improvement_rate <= low_improvement_threshold or non_improving_streak >= stagnation_threshold - 1) and edge_length_skew <= 0.62: switch to random_like_heuristic."}, {"phase": "extreme_skew_protection", "rule": "if edge_length_skew > 0.66: do not switch heuristics; keep current heuristic for stability."}], "stagnation_switch_heuristic": "nearest_neighbor_multistart", "stagnation_threshold": 3, "temperature_scale": 1.12, "time_budget_trigger": 0.65, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}]

phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.120261, selector regret 0.048001, runtime-adjusted gap 0.634817.
- Family transfer 0.001857 and combined transfer 0.045479.
- Runtime 7723.547714 ms, runtime inflation 4.278659, mean novelty 0.722823, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: annealed_multi_start:cluster_first_local_search:farthest_insertion_two_opt:annealed_multi_start:stag3:cand-3:restart1.
- Controller rule summary: {"acceptance_bias": 0.0, "candidate_limit_offset": -3, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic":

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"farthest_insertion_two_opt", "description": "Deterministic instance-adaptive hyper-heuristic controller selecting from a frozen portfolio using structure cues (cluster/grid/bottleneck/corridor/trap) and conservative online switching under stagnation and perturbation failure; biases toward farthest-insert", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.12, "name": "deterministic_instance_adaptive_tsp_hh_v5", "nearest_neighbor_trap_heuristic": "cheapest_insertion_two_opt", "random_like_heuristic": "annealed_multi_start", "restart_offset": 1, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.25, "time_budget_trigger": 0.6, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}
```

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.07918, selector regret 0.00692, runtime-adjusted gap 0.07918.
- Family transfer 0.010298 and combined transfer 0.035675.
- Runtime 509.750157 ms, runtime inflation -0.651611, mean novelty 0.548573, and complexity 0.76.
- Pareto efficient: True.

Judge Appendix

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TSP Adaptive Heuristic Portfolio Suite - Conservative Result Interpretation

Primary Endpoint: Held-out TSPLIB Gap

- The best held-out TSPLIB gap (0.07226) is achieved by the **best single fixed heuristic** ("farthest_insertion_two_opt").
- The **oracle selector** achieves the same TSPLIB gap (0.07226), marking this as a performance upper bound.
- The **supervised ML selector** also reaches 0.07226 gap but with slightly higher runtime inflation.
- Adaptive controllers yield higher TSPLIB gaps (e.g., 0.214 for the "llm_evolved_adaptive_controller") than the best fixed heuristic, indicating performance degradation from adaptive selection on this metric.

Selector Regret vs Oracle Portfolio

- Best single fixed heuristic and supervised ML selectors show zero selector regret, matching the oracle on TSPLIB gap.
- Adaptive controllers show notable selector regret (0.142 for "llm_evolved_adaptive_controller"), implying suboptimal heuristic choices compared to oracle.
- Random portfolios show the highest selector regret (0.117 initially), which aligns with poor TSPLIB gap results.

Runtime-Adjusted Gap & Runtime Inflation

- The best fixed heuristic and oracle selector show similar runtime-adjusted gaps (~ 0.073) with negligible runtime inflation.
- Adaptive controllers reduce runtime (runtime inflation negative, e.g. -0.73 for "llm_evolved_adaptive_controller") but at cost of increased gap (0.214).
- One adaptive variant ("llm_evolved_controller_diversity_failure_replay") shows large runtime inflation (4.28) and worse adjusted gap (0.635), which is not Pareto efficient.
- The static selector ("llm_static_selector") runs very fast (273 ms, much lower than ~ 1500 ms for fixed heuristic) but incurs higher gap (0.229) and negative runtime inflation (-0.81), showing efficiency in speed but worse solution quality.

Pareto Efficiency and Complexity

- Pareto efficient conditions include:
- Best single fixed heuristic (simple, no adaptation, best gap)
- Full solver evolution (moderate gap 0.079, reduced runtime)
- LLM evolved adaptive controller (higher gap but substantial runtime reduction)
- LLM static selector (fastest runtime, higher gap)
- Adaptive controllers have moderate complexity (0.62) and interpretable tuning rules, indicating structured instance-adaptive control.

Cross-Family Transfer Performance

- Transfer gaps are lowest for oracle selector (0.0267) and best single heuristic (0.0312), supporting robust cross-family transfer.
- Adaptive controllers show higher transfer gap (0.107 - 0.127), indicating some loss of generality or robustness.
- Random portfolio and static selector have higher transfer gaps, consistent with poorer control policies.

Summary and Recommendations

- ****Best single fixed heuristic ("farthest_insertion_two_opt") remains the strongest performer on held-out TSPLIB gap, matching oracle selector without runtime overhead.****
- ****Adaptive selectors, especially the evolved LLM adaptive controller, provide notable runtime savings (roughly 2-4x speedup) but currently at a significant cost in solution quality (larger TSPLIB gap and selector regret).****
- The ****static LLM-based selector is very fast but yields high gap****, making it Pareto efficient only where runtime dominates.
- Adaptive controllers demonstrate interpretable instance-adaptive scheduling with well-defined selection and tuning rules; however, their gap-to-runtime tradeoff suggests further tuning or hybrid strategies may be needed.
- Cross-family transfer favors fixed heuristics and oracle-like selectors; adaptive controllers show some vulnerability here.
- No new algorithm discoveries are evident; improvements stem from hyper-heuristic control over known heuristics.

- Deployment should prioritize either best fixed heuristic for quality or adaptive controllers if runtime constraints dominate and some quality loss is acceptable.

Interpretation Priorities Addressed:

Criterion	Best Single Fixed	Oracle Selector	LLM Evolved Adaptive Controller	LLM Static Selector	Random Portfolio
Held-out TSPLIB gap	**0.07226 (best)**	0.07226	0.214 (worse)	0.229 (worse)	0.189 (worse)
Selector regret vs oracle	0	0 (oracle)	0.142 (notable regret)	0.157 (high regret)	0.117
Runtime-adjusted gap	~0.072	0.0736	0.214 (higher)	0.229 (higher)	0.193
Runtime inflation	0	+0.019	−0.73 (faster)	−0.81 (faster)	+0.02
Pareto efficiency	Yes	No	Yes	Yes	No
Cross-family transfer gap	0.031	0.027	0.107	0.127	0.094

Final Conservative Conclusion

- The frozen best single heuristic presents a baseline with the best held-out performance and no runtime inflation.
- The oracle selector validates this baseline as an upper bound but is not deployable.
- The adaptive LLM evolved controller shows promise in runtime savings with clear instance-adaptive control but needs improvement to approach oracle performance without excessive gap.
- Static and random portfolios underperform on gap and have weaker transfer.
- Future work should focus on improving instance-adaptive selection accuracy and preserving solution quality while maintaining runtime benefits.